

# Daily Geometry Drill #5

Sheet by Henry Koduthore

| Section | Topic                    | Questions | Target time  |
|---------|--------------------------|-----------|--------------|
| A       | Rapid Recognition        | 10        | 2 min 30 sec |
| B       | Manipulation Drills      | 10        | 8 min        |
| C       | Substitution & Structure | 8         | 10 min       |
| D       | Challenge                | 5         | 15 min       |

*New toolkit today: Angle at the centre · Cyclic quadrilaterals · Tangent  $\perp$  radius · Power of a point · Touching circles: join the centres.*

*Attempt each question before checking answers. Time yourself. No calculators. Diagrams are not to scale.*

## Section A Rapid Recognition

( $\leq 15$  s each)

*Angle at the centre, angles on the same arc, cyclic opposite angles, the alternate segment, and power of a point. Know them cold.*

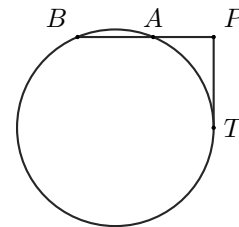
- In a circle with centre  $O$ , the angle at the circumference subtending arc  $AB$  is  $20^\circ$ . Write down the angle at the centre.
- $AB$  is a diameter of a circle and  $P$  is any other point on the circle. Write down  $\angle APB$ .
- $ABCD$  is a cyclic quadrilateral with  $\angle A = 70^\circ$  and  $\angle B = 85^\circ$ . Write down  $\angle C$ .
- The tangent at  $T$  to a circle makes an angle of  $40^\circ$  with the chord  $TA$ . By the alternate segment theorem, write down the angle in the alternate segment subtended by  $TA$ .
- Chords  $AB$  and  $CD$  of a circle intersect inside at  $P$ , with  $PA = 2$ ,  $PB = 6$  and  $PC = 3$ . Write down  $PD$ .
- A tangent from  $P$  touches the circle at  $T$ , and a secant through  $P$  cuts the circle at  $A$  then  $B$ . Given  $PT = 3$  and  $PA = 1$ , write down  $PB$ .
- Write down the length of the tangent from  $(7, 0)$  to  $x^2 + y^2 = 36$ .
- Two tangents from the point  $P$  to a circle of unknown radius make an angle of  $60^\circ$  and  $OP = 6$ . Write down the length of each tangent.
- The two tangents from  $P$  to the circle  $x^2 + y^2 = 25$  make an angle of  $120^\circ$ . Write down  $OP$ .
- In the cyclic quadrilateral  $ABCD$ ,  $\angle BAC = 30^\circ$ . Write down  $\angle BDC$ .

## Section B Manipulation Drills

( $\leq 50$  s each)

*Draw the radius to the point of contact: it is perpendicular to the tangent. Then look for a right angle or a power of a point.*

1. A tangent at  $T$  makes an angle of  $55^\circ$  with chord  $TA$ . The angle in the alternate segment subtended by  $TA$  is?
  - A)  $35^\circ$
  - B)  $55^\circ$
  - C)  $110^\circ$
  - D)  $70^\circ$
2.  $ABCD$  is cyclic with  $\angle A = 2x$  and  $\angle C = 3x$ . Find  $x$ .
  - A) 24
  - B) 30
  - C) 36
  - D) 72
3. A *tangential* (circumscribed) quadrilateral  $ABCD$  has  $AB = 5$ ,  $BC = 7$  and  $CD = 6$ . By Pitot's theorem ( $AB + CD = BC + DA$ ), find  $DA$ .
  - A) 4
  - B) 6
  - C) 8
  - D) 18
4. From external  $P$ , tangent  $PT = 12$  touches a circle and secant  $PAB$  meets it with  $PA = 8$ . Find the chord  $AB$ .
  - A) 6
  - B) 10
  - C) 12
  - D) 18
5. From an external point  $P$ , two secants meet the same circle at  $P-A-B$  and  $P-C-D$ , with  $PA = 4$ ,  $PB = 18$  and  $PC = 6$ . Write down the length of the chord  $CD$ .
6. The point  $P$  has  $OP = 6$  with respect to a circle of radius 3. The angle between the two tangents from  $P$  is?
  - A)  $30^\circ$
  - B)  $45^\circ$
  - C)  $60^\circ$
  - D)  $90^\circ$
7. Two circles of radius 6 each pass through the other's centre. They meet at  $A$  and  $B$ . How long is  $AB$ ? (after *Primer 1.4 P16*)
  - A) 6



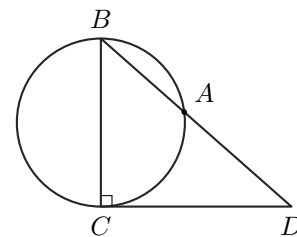
- B) 12  
 C)  $6\sqrt{2}$   
 D)  $6\sqrt{3}$   
 E)  $3\sqrt{3}$
8. Two concentric circles have radii in the ratio 2 : 5.  $AC$  is a diameter of the larger circle, and the chord  $BC$  of the larger circle touches the smaller circle. If  $AB = 8$ , what is the radius of the larger circle? (after Primer 1.4 P13)
- A) 16  
 B) 20  
 C) 8  
 D) 12  
 E) 10
9. The line  $ST$  is a chord of the larger of two concentric circles and touches the smaller one. If  $ST = 30$ , find the area of the region between the circles. (after Primer 1.4 P8)
10.  $AB$  is a diameter of a circle and  $C$  is a point on the circle with  $\angle ABC = 55^\circ$ . Write down  $\angle BAC$ .

### Section C Substitution & Structure

( $\leq 90$  s each)

SMC Q18–22 level. Join the centres; tangent circles have centres  $r_1 + r_2$  apart and a common point on that line.

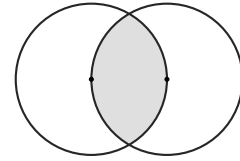
1.  $BC$  is a diameter of a circle and  $CD$  is the tangent at  $C$ . The line  $BD$  meets the circle again at  $A$ . If  $AB = 7$  and  $AD = 9$ , what is the radius of the circle? (after Primer 1.4 P40)
- A) 12  
 B)  $4\sqrt{7}$   
 C)  $2\sqrt{7}$   
 D) 8  
 E) 6



2. A regular octagon is inscribed in a circle of radius 2, and  $X$  is any point on the circle. What is the sum of the squares of the distances from  $X$  to the eight vertices? (after SMC 2025 Q21)
- A) 64  
 B) 32  
 C) 128  
 D) 48  
 E) 16

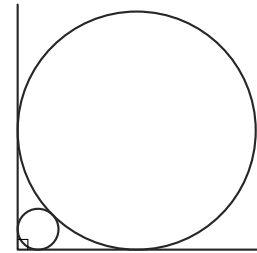
3. Two circles of radius 2 each pass through the other's centre. What is the area of the shaded region common to both circles? (after Primer 1.4 P24)

- A)  $\frac{4\pi}{3} - \sqrt{3}$   
 B)  $\frac{8\pi}{3} - \sqrt{3}$   
 C)  $\frac{4\pi}{3} + 2\sqrt{3}$   
 D)  $\frac{8\pi}{3} - 2\sqrt{3}$   
 E)  $2\pi - 2\sqrt{3}$



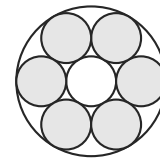
4. A circle of radius 1 touches both arms of a right angle. A larger circle touches both arms and the first circle. What is the radius of the larger circle? (after Primer 1.4 P35)

- A)  $1 + \sqrt{2}$   
 B)  $2 + \sqrt{2}$   
 C)  $2\sqrt{2}$   
 D) 3  
 E)  $3 + 2\sqrt{2}$



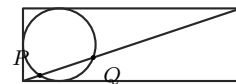
5. Six equal discs lie between two concentric circles. Each disc touches its two neighbours and both circles. The inner circle has radius 1. What is the radius of the outer circle? (after SMC 2016 Q21)

- A) 3  
 B)  $2 + \sqrt{3}$   
 C)  $1 + \sqrt{3}$   
 D)  $2\sqrt{3}$   
 E) 4



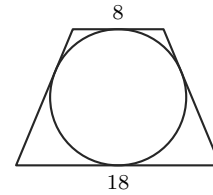
6. A circle of radius 1 touches three sides of a  $6 \times 2$  rectangle. A diagonal of the rectangle cuts the circle at  $P$  and  $Q$ . What is the length  $PQ$ ? (after SMC 2017 Q19)

- A)  $\frac{4\sqrt{5}}{5}$   
 B)  $\frac{2\sqrt{15}}{5}$   
 C)  $\sqrt{3}$   
 D)  $\frac{6}{5}$   
 E)  $\frac{2\sqrt{10}}{5}$



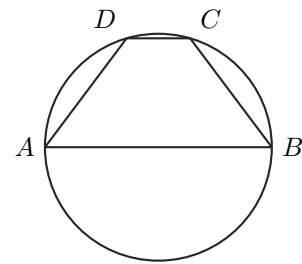
7. An isosceles trapezium has parallel sides of lengths 8 and 18, and all four of its sides touch a circle. What is the radius of the circle? (after SMC 2012 Q20)

- A)  $\frac{13}{2}$   
 B) 5  
 C) 12  
 D) 6  
 E)  $4\sqrt{3}$



8.  $ABCD$  is a cyclic quadrilateral in which  $AB$  is a diameter of the circle,  $AB = 25$  and  $AD = BC = 15$ . What is the length  $CD$ ? (after Primer 1.4 P33)

- A) 10  
 B) 15  
 C) 13  
 D) 9  
 E) 7



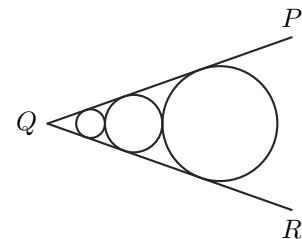
### Section D Challenge

(SMC Q23–25 / BMO1 difficulty)

*Coordinates, the sine rule or a chain of similar figures. Diagrams are not to scale.*

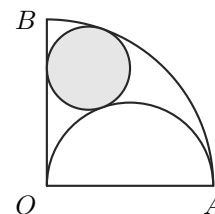
1. Three circles each touch both arms  $QP$  and  $QR$  of an angle, and each touches the next. The largest radius is 4 times the smallest. What is  $\cos \angle PQR$ ? (after SMC 2011 Q24)

- A)  $\frac{7}{9}$   
 B)  $\frac{1}{3}$   
 C)  $\frac{2}{3}$   
 D)  $\frac{8}{9}$   
 E)  $\frac{4\sqrt{2}}{9}$



2.  $OAB$  is a quarter circle with radius 4. A semicircle is drawn on  $OA$  as diameter, inside the quarter circle. A circle touches  $OB$ , the semicircle and the arc  $AB$ . What is its radius? (after SMC 2014 Q19)

- A)  $\frac{4}{3}$   
 B) 1  
 C)  $\frac{2}{3}$   
 D)  $\sqrt{2}$   
 E)  $\frac{3}{4}$



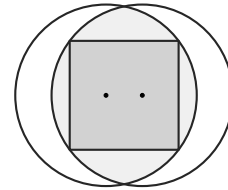
3. A triangle with angles  $30^\circ$ ,  $45^\circ$  and  $105^\circ$  is inscribed in a circle of radius 2. What is its area?

(after SMC 2021 Q22)

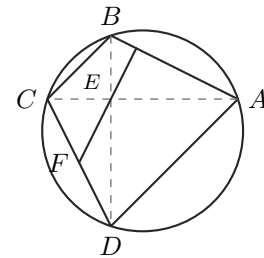
- A)  $2 + \sqrt{3}$
- B)  $2\sqrt{3}$
- C)  $1 + \sqrt{3}$
- D)  $3 + \sqrt{3}$
- E) 2

4. Two circles of radius 5 have centres 2 apart. A square is inscribed in the region common to both circles, with its sides parallel and perpendicular to the line of centres. What is the area of the square? (after SMC 2019 Q25)

- A) 64
- B) 32
- C) 25
- D) 36
- E) 48



5.  $ABCD$  is a cyclic quadrilateral whose diagonals are perpendicular and meet at  $E$ . The line through  $E$  perpendicular to  $AB$  meets  $CD$  at  $F$ . Prove that  $F$  is the midpoint of  $CD$ .




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“With circles, draw what you cannot see: the centres, the radii to the points of contact, and the line joining two centres.”

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Found a mistake or want to contribute a sheet?

Visit the open-source project at [github.com/The-CerealDev/speed-maths](https://github.com/The-CerealDev/speed-maths)